LOG

Analisi risultati bonus 1

Signal 0 Slow

```
--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:-1.11 (SENT)
> LoRaWAN Uplink Scheduled  : 25 s
> Saved Data Volume         : 9 bytes
> Execution Time per window : 3075 us
> End-to-End Latency        : 1273 ms
> FFT Dominant Frequency    : 0.55 Hz
> Adaptive Sampling Freq    : 10.0 Hz
> Average    : -1.11
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:-1.11 (SENT)
> LoRaWAN Uplink Scheduled  : 20 s
> Saved Data Volume         : 9 bytes
> Execution Time per window : 3075 us
> End-to-End Latency        : 1273 ms
> FFT Dominant Frequency    : 0.55 Hz
> Adaptive Sampling Freq    : 10.0 Hz
> Average Calculated    : -1.11
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:0.12 (SENT)
> LoRaWAN Uplink Scheduled  : 15 s
> Saved Data Volume         : 8 bytes
> Execution Time per window : 3038 us
> End-to-End Latency        : 12703 ms
> FFT Dominant Frequency    : 5.00 Hz
> Adaptive Sampling Freq    : 10.0 Hz
> Average Calculated    : 0.12
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:0.12 (SENT)
> LoRaWAN Uplink Scheduled  : 10 s
> Saved Data Volume         : 8 bytes
> Execution Time per window : 3038 us
> End-to-End Latency        : 12703 ms
```

```
> FFT Dominant Frequency    : 5.00 Hz
> Adaptive Sampling Freq    : 10.0 Hz
> Average Calculated    : 0.12
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:-0.10 (SENT)
> LoRaWAN Uplink Scheduled  : 5 s
> Saved Data Volume         : 9 bytes
> Execution Time per window : 3043 us
> End-to-End Latency        : 10163 ms
> FFT Dominant Frequency    : 4.00 Hz
> Adaptive Sampling Freq    : 8.0 Hz
> Average Calculated    : -0.10
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:-0.10 (SENT)
> LoRaWAN Transmission      : -0.10 (SENT TO TTN)
> Saved Data Volume         : 9 bytes
> Execution Time per window : 3043 us
> End-to-End Latency        : 10163 ms
> FFT Dominant Frequency    : 4.00 Hz
> Adaptive Sampling Freq    : 8.0 Hz
> Average Calculated    : -0.10
-----------------------------------
```

Transizione (Segnale Veloce)

Signal 1 Fast

```
--- [SYSTEM PERFORMANCE REPORT] ---

> MQTT Transmission         : Avg:-0.10 (SENT)

> LoRaWAN Uplink Scheduled  : 25 s

> Saved Data Volume         : 9 bytes

> Execution Time per window : 3043 us

> End-to-End Latency        : 10163 ms

> FFT Dominant Frequency    : 4.00 Hz

> Adaptive Sampling Freq    : 8.0 Hz
```

```
> Average Calculated    : -0.10
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:0.01 (SENT)
> LoRaWAN Uplink Scheduled  : 20 s
> Saved Data Volume         : 8 bytes
> Execution Time per window : 3070 us
> End-to-End Latency        : 12576 ms
> FFT Dominant Frequency    : 4.94 Hz
> Adaptive Sampling Freq    : 9.92 Hz
> Average Calculated    : 0.01
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:0.01 (SENT)
> LoRaWAN Uplink Scheduled  : 15 s
> Saved Data Volume         : 8 bytes
> Execution Time per window : 3070 us
> End-to-End Latency        : 12576 ms
> FFT Dominant Frequency    : 4.94 Hz
> Adaptive Sampling Freq    : 9.92 Hz
> Average Calculated    : 0.01
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
```

> MQTT Transmission : Avg:-0.02 (SENT)

> LoRaWAN Uplink Scheduled : 10 s

> Saved Data Volume : 9 bytes

> Execution Time per window : 3047 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 44.29 Hz

> Adaptive Sampling Freq : 88.58 Hz

> Average Calculated : -0.02

--- [SYSTEM PERFORMANCE REPORT] ---

> MQTT Transmission : Avg:0.00 (SENT)

> LoRaWAN Uplink Scheduled : 5 s

> Saved Data Volume : 8 bytes

> Execution Time per window : 3049 us

> End-to-End Latency : 1146 ms

> FFT Dominant Frequency : 18.13 Hz

> Adaptive Sampling Freq : 36.26 Hz

> Average Calculated : 0.00

--- [SYSTEM PERFORMANCE REPORT] ---

> MQTT Transmission : Avg:-0.00 (SENT)

> LoRaWAN Transmission : -0.00 (SENT TO TTN)

> Saved Data Volume : 9 bytes

> Execution Time per window : 3059 us

> End-to-End Latency : 1146 ms

> FFT Dominant Frequency : 18.13 Hz

> Adaptive Sampling Freq : 36.26 Hz

> Average Calculated : -0.00

Transizione (Segnale Complesso)

Signal 2 Complex

--- [SYSTEM PERFORMANCE REPORT] ---

> MQTT Transmission : Avg:-0.03 (SENT)

> LoRaWAN Uplink Scheduled : 25 s

> Saved Data Volume : 9 bytes

> Execution Time per window : 3041 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 11.07 Hz

> Adaptive Sampling Freq : 22.14 Hz

> Average Calculated : -0.03

--- [SYSTEM PERFORMANCE REPORT] ---

> MQTT Transmission : Avg:-0.04 (SENT)

> LoRaWAN Uplink Scheduled : 20 s

> Saved Data Volume : 9 bytes

> Execution Time per window : 3058 us

> End-to-End Latency : 2797 ms

```
> FFT Dominant Frequency    : 11.07 Hz
> Adaptive Sampling Freq    : 22.14 Hz
> Average Calculated    : -0.04
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:-0.01 (SENT)
> LoRaWAN Uplink Scheduled  : 15 s
> Saved Data Volume         : 9 bytes
> Execution Time per window : 3045 us
> End-to-End Latency        : 4575 ms
> FFT Dominant Frequency    : 18.13 Hz
> Adaptive Sampling Freq    : 36.26 Hz
> Average Calculated    : -0.01
-----------------------------------

--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:0.05 (SENT)
> LoRaWAN Uplink Scheduled  : 10 s
> Saved Data Volume         : 8 bytes
> Execution Time per window : 3052 us
> End-to-End Latency        : 2797 ms
> FFT Dominant Frequency    : 11.07 Hz
> Adaptive Sampling Freq    : 22.14 Hz
> Average Calculated    : 0.05
-----------------------------------
```

```
--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:-0.01 (SENT)
> LoRaWAN Uplink Scheduled  : 5 s
> Saved Data Volume         : 9 bytes
> Execution Time per window : 3042 us
> End-to-End Latency        : 2797 ms
> FFT Dominant Frequency    : 11.07 Hz
> Adaptive Sampling Freq    : 22.14 Hz
> Average Calculated    : -0.01
-----------------------------------
```

```
--- [SYSTEM PERFORMANCE REPORT] ---
> MQTT Transmission         : Avg:0.01 (SENT)
> LoRaWAN Transmission      : 0.01 (SENT TO TTN)
> Saved Data Volume         : 8 bytes
> Execution Time per window : 3044 us
> End-to-End Latency        : 4575 ms
> FFT Dominant Frequency    : 18.13 Hz
> Adaptive Sampling Freq    : 36.26 Hz
> Average Calculated    : 0.01
-----------------------------------
```

Analisi Bonus parte 2

Test 1.1: Bassa Probabilità di Anomalia (p = 1%)

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3496 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 11.07 Hz

> Adaptive Sampling Freq : 22.14 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 279 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 100.0 % / 0.0 %

> Final Average (Z-Score) : -0.10

> Final Average (Hampel) : -0.10

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3499 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 11.07 Hz

> Adaptive Sampling Freq : 22.14 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 279 us

> Z-Score (TPR/FPR) : 66.7 % / 0.0 %

> Hampel (TPR/FPR) : 66.7 % / 0.0 %

> Final Average (Z-Score) : -0.10

> Final Average (Hampel) : -0.10

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3479 us

> End-to-End Latency : 4575 ms

> FFT Dominant Frequency : 18.14 Hz

> Adaptive Sampling Freq : 36.28 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 278 us

> Z-Score (TPR/FPR) : 40.0 % / 0.0 %

> Hampel (TPR/FPR) : 40.0 % / 0.0 %

> Final Average (Z-Score) : 0.04

> Final Average (Hampel) : 0.03

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3513 us

> End-to-End Latency : 2798 ms

> FFT Dominant Frequency : 11.06 Hz

> Adaptive Sampling Freq : 22.12 Hz

> CPU Time Z-Score : 198 us

> CPU Time Hampel : 297 us

> Z-Score (TPR/FPR) : 50.0 % / 0.0 %

> Hampel (TPR/FPR) : 50.0 % / 0.0 %

> Final Average (Z-Score) : 0.14

> Final Average (Hampel) : 0.14

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3488 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 11.06 Hz

> Adaptive Sampling Freq : 22.12 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 274 us

> Z-Score (TPR/FPR) : 33.3 % / 0.0 %

> Hampel (TPR/FPR) : 33.3 % / 0.0 %

> Final Average (Z-Score) : -0.12

> Final Average (Hampel) : -0.12

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3501 us

> End-to-End Latency : 4576 ms

> FFT Dominant Frequency : 18.13 Hz

> Adaptive Sampling Freq : 36.26 Hz

> CPU Time Z-Score : 198 us

> CPU Time Hampel : 282 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 100.0 % / 0.0 %

> Final Average (Z-Score) : 0.05

> Final Average (Hampel) : 0.05

Test 1.2: Media Probabilità di Anomalia (p = 5%)

Bonus

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 3476 us

> End-to-End Latency : 1274 ms

> FFT Dominant Frequency : 5.04 Hz

> Adaptive Sampling Freq : 10.1 Hz

> Hardware Max Sampling Freq: 16633 Hz

> CPU Time Z-Score : 198 us

> CPU Time Hampel : 272 us

> Z-Score (TPR/FPR) : 50.0 % / 0.0 %

> Hampel (TPR/FPR) : 50.0 % / 0.0 %

> Final Average (Z-Score) : 0.19

> Final Average (Hampel) : 0.19

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 3476 us

> End-to-End Latency : 1274 ms

> FFT Dominant Frequency : 5.04 Hz

> Adaptive Sampling Freq : 10.1 Hz

> Hardware Max Sampling Freq: 16633 Hz

> CPU Time Z-Score : 198 us

> CPU Time Hampel : 272 us

> Z-Score (TPR/FPR) : 50.0 % / 0.0 %

> Hampel (TPR/FPR) : 50.0 % / 0.0 %

> Final Average (Z-Score) : 0.19

> Final Average (Hampel) : 0.19

> CPU Execution Time : 3503 us

> End-to-End Latency : 1273 ms

> FFT Dominant Frequency : 5.09 Hz

> Adaptive : 10.2 Hz

> Hardware Max Sampling Freq: 16633 Hz

> CPU Time Z-Score : 196 us

> CPUZ-Score (TPR/FPR) : 16.7 % / 0.0 %

> Hampel (TPR/FPR) : 41.7 % / 0.0 %

> Final Average (Z-Score) : 0.15

> Final Average (Hampel) : -0.15

SYSTEM PERFOR

> CPU Execution Time : 3503 us

> End-to-End Latency : 1273 ms

> FFT Dominant Frequency : 5.09 Hz

> Adaptive Sampling Freq : 10.2 Hz

> Hardware Max Sampling Freq: 16633 Hz

> CPU Time Z-Score : 196 us

> CPU Time Hampel : 285 u : 16.7 % / 0.0 %

> Hampel (TPR/FPR) : 41.7 % / 0.0 %

> Final Average (Z-Score) : 0.15

> Final Average (Hampel) : -0.15

```
-----------------------------------

SYSTEM PERFORMANCE REPORT
> CPU Execution Time        : 3503 us
> End-to-End Latency        : 1273 ms
> FFT Dominant Frequency    : 5.09 Hz
> Adaptive Sampling Freq    : 10.2 Hz
> Hardware Max Sampling Freq: 16633 Hz
> CPU Time Z-Score          : 196 us
> CPU Time Hampel           : 285 us
> Z-Score (TPR/FPR)         : 16.7 % / 0.0 %
> Hampel (TPR/FPR)          : 41.7 % / 0.0 %
> Final Average (Z-Score)   : 0.15
> Final Average (Hampel)    : -0.15
-----------------------------------

SYSTEM PERFORMANCE REPORT
> CPU Execution Time        : 3493 us
> End-to-End Latency        : 1273 m    : 10.0 Hz
> Hardware Max Sampling Freq: 16633 Hz
> CPU Time  Z-Score (TPR/FPR)         : 0.0 % / 0.0 %
> Hampel (TPR/FPR)          : 0.0 % / 0.0 %
> Final Average (Z-Score)   : -0.17
> Final Average (Hampel)    : -0.17
-----------------------------------
```

Test 1.3: Alta Probabilità di Anomalia (p = 10%)

Bonus

> End-to-End Latency : 1275 ms

> FFT Dominant Frequency : 4.99 Hz

> Adaptive Sampling Freq : 10.0 Hz

> Hardware Max Sampling Freq: 16640 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 275 us

> Z- : 7.1 % / 0.0 %

> Hampel (TPR/FPR) : 21.4 % / 0.0 %

> Final Average (Z-Score) : -0.43

> Final Average (Hampel) : -0.21

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 3477 us

> End-to-End Latency : 1275 ms

: 10.0 Hz

> Hardware Max Sampling Freq: 16640 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 275 us

> Z-Score (TPR/FPR) : 7.1 % / 0.0 %

> Hampel (TPR/FPR) : 21.4 % / 0.0 %

> Final Average (Z-Score) : -0.43

> Final Average (Hampel) : -0.21

SYSTEM PERFORMs

> End-to-End Latency : 1273 ms

> FFT Dominant Frequency : 4.93 Hz

> Adaptive Sampling Freq : 10.0 Hz

> Hardware Max Sampling Freq: 16640 Hz

> CPU Time Z-Score : 200 us

> CPU Time Hampel : 275 us

> Z-Score (TPR/FPR) : 27.3 % / 0.0 %

> Hampel (TPR/FPR) : 27.3 % / 0.0 %

> Final Average (Z-Score) : -0.29

> Final Average (Hampel) : -0.29

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 3488 us

> End-to-End Latency : 1273 ms

> FFT Dominant Frequency : 4.93 Hz

> Adaptive Sampling Freq : 10.0 Hz

> Haeq: 16640 Hz

> CPU Time Z-Score : 200 us

> CPU R) : 27.3 % / 0.0 %

> Hampel (TPR/FPR) : 27.3 % / 0.0 %

> Final Average (Z-Score) : -0.29

> Final Average (Hampel) : -0.29

SYSTRFORMANCE REPORT

> CPU Execution Time : 3488 us

> End-to-End Latency : 1273 ms

> FFT Dominant Frequency : 4.93 Hz

> Adaptive Sampling Freq Score : 200 us

> CPU Time Hampel : 275 us

> Z-Score (TPR/FPR) : 27.3 % / 0.0 %

> Hampel (TPR/FPR) : 27.3 % / 0.0 %

> Final Average (Z-Score) : -0.29

> Final Average (Hampel) : -0.29

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 3480 us

> End-to-End Latency : 1273 ms

> FFT Dominant Frequency : 4.95 Hz

> Adaptive Sampling Freq : 10.0 Hz

> Hardware Max Sampling Freq: 16640 Hz

> CPU Time Z-Score : 198 us

> CPU Time Hampel : 278 us

> Z-Score (TPR/FPR) : 23.1 % / 0.0 %

> Hampel (TPR/FPR) : 23.1 % / 0.0 %

> Final Average (Z-Score) : -0.19

> Final Average (Hampel) : -0.20

Analisi dell'Esperimento 2: Variazione della dimensione della finestra

Test 2.1: Finestra Piccola (N = 64)

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 1799 us

> End-to-End Latency : 632 ms

> FFT Dominant Frequency : 4.83 Hz

> Adaptive Sampling Freq : 10.0 Hz

> Hardware Max Sampling Freq: 16497 Hz

> CPU Time Z-Score : 106 us

> CPU Time Hampel : 147 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 100.0 % / 0.0 %

> Final Average (Z-Score) : -0.16

> Final Average (Hampel) : -0.16

SYSTEM PERFORMANCE REPORT

End-to-End Latency : 632 ms

> FFT Dominant Frequency : 4.97 Hz

> Adaptive Sampling Freq : 10.0 Hz

> Hardware Max Sampling Freq: 16497 Hz

> CPU Time Z-Score : 107 us

> CPU Time Hampel : 148 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 0.0 % / 0.0 %

> Final Average (Z-Score) : -0.17

> Final Average (Hampel) : -0.37

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 1728 us

> End-to-End Latency : 1010 ms

> FFT Dominant Frequency : 7.83 Hz

> Adaptive Sampling Freq : 15.7 Hz

> Hardware Max Sampling Freq: 16497 Hz

> CPU Time Z-Score : 106 us

> CPU Time Hampel : 152 us

> Z-Score (TPR/FPR) : 0.0 % / 0.0 %

> Hampel (TPR/FPR) : 0.0 % / 0.0 %

> Final Average (Z-Score) : 0.03

> Final Average (Hampel) : 0.03

> CPU Execution Time : 1715 us

> End-to-End Latency : 947 ms

> FFT Dominant Frequency : 7.59 Hz

> Adaptive Sampling Freq : 15.2 Hz

> Hardware Max Sampling Freq: 16497 Hz

> CPU Time Z-Score : 106 us

> CPU Time Hampel : 147 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 100.0 % / 0.0 %

> Final Average (Z-Score) : -0.22

> Final Average (Hampel) : -0.22

SYSTEM PERFORMANCE REPO

> CPU Execution Time : 1734 us

> End-to-End Latency : 947 ms

> FFT Do: 7.54 Hz

> Adaptive Sampling Freq : 15.1 Hz

> Hardware Max Sampling Freq: 16497 Hz

> CPU Time Z-Score : 105 us

> CPU Time Hampel : 151 us

> Z-Score (TPR/FPR) : 100.0inal Average (Z-Score) : 0.29

> Final Average (Hampel) : 0.29

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 1721 us

> End-to-End Latency : 884 ms

> FFT Dominant Frequency : 7.28 Hz

> Adaptive Sampling Freq : 14.6 Hz

> Hardware Max Sampling Freq: 16497 Hz

> CPU Time Z-Score : 106 us : 146 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 100.0 % / 0.0 %

> Final Average (Z-Score) : -0.09

> Final Average (Hampel) : -0.09

Test 2.2: Finestra Standard (N = 128)

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3496 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 11.07 Hz

> Adaptive Sampling Freq : 22.14 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 279 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 100.0 % / 0.0 %

> Final Average (Z-Score) : -0.10

> Final Average (Hampel) : -0.10

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3499 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 11.07 Hz

> Adaptive Sampling Freq : 22.14 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 279 us

> Z-Score (TPR/FPR) : 66.7 % / 0.0 %

> Hampel (TPR/FPR) : 66.7 % / 0.0 %

> Final Average (Z-Score) : -0.10

> Final Average (Hampel) : -0.10

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3479 us

> End-to-End Latency : 4575 ms

> FFT Dominant Frequency : 18.14 Hz

> Adaptive Sampling Freq : 36.28 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 278 us

> Z-Score (TPR/FPR) : 40.0 % / 0.0 %

> Hampel (TPR/FPR) : 40.0 % / 0.0 %

> Final Average (Z-Score) : 0.04

> Final Average (Hampel) : 0.03

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3513 us

> End-to-End Latency : 2798 ms

> FFT Dominant Frequency : 11.06 Hz

> Adaptive Sampling Freq : 22.12 Hz

> CPU Time Z-Score : 198 us

> CPU Time Hampel : 297 us

> Z-Score (TPR/FPR) : 50.0 % / 0.0 %

> Hampel (TPR/FPR) : 50.0 % / 0.0 %

> Final Average (Z-Score) : 0.14

> Final Average (Hampel) : 0.14

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3488 us

> End-to-End Latency : 2797 ms

> FFT Dominant Frequency : 11.06 Hz

> Adaptive Sampling Freq : 22.12 Hz

> CPU Time Z-Score : 197 us

> CPU Time Hampel : 274 us

> Z-Score (TPR/FPR) : 33.3 % / 0.0 %

> Hampel (TPR/FPR) : 33.3 % / 0.0 %

> Final Average (Z-Score) : -0.12

> Final Average (Hampel) : -0.12

--- [SYSTEM PERFORMANCE REPORT] ---

> CPU Execution Time : 3501 us

> End-to-End Latency : 4576 ms

> FFT Dominant Frequency : 18.13 Hz

> Adaptive Sampling Freq : 36.26 Hz

> CPU Time Z-Score : 198 us

> CPU Time Hampel : 282 us

> Z-Score (TPR/FPR) : 100.0 % / 0.0 %

> Hampel (TPR/FPR) : 100.0 % / 0.0 %

> Final Average (Z-Score) : 0.05

> Final Average (Hampel) : 0.05

Test 2.3: Finestra Grande (N = 256)

SYSTEM PE> CPU Execution Time : 7268 us

> End-to-End Latency : 2558 ms

> FFT Dominant Frequency : 5.04 Hz

> Adaptive Sampling Freq : 10.1 Hz

> Hardware Max Sampling Freq: 16629 H2 us

> CPU Time Hampel : 557 us

> Z-Score (TPR/FPR) : 20.0 % / 0.0 %

> Hampel (TPR/FPR) : 20.0 % / 0.0 %

> Final Average (Z-Score) : 0.07

> Final Average (Hampel) : 0.07

SYSTEM PER

> CPU Execution Time : 7268 us

> End-to-End Latency : 2558 ms

> Adaptive Sampling Freq : 10.1 Hz

> Hardware Max Sampling Freq: 16629 Hz

> CPU Time Z-Score : 382 us

> CPU Time Hampel : 557 us

> Z-Score (TPR/FPR) : 20.0 % / 0.0 %

> Hampel (TPR/FPR) : 20.0 % / 0.0 %

> Final Average (Z-Score) : 0.07

> Final Average (Hampel) : 0.07

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 7268 us

> End-to-End Latency : 2558 ms

> FFT Dominant Frequency : 5.04 Hz

> Adaptive Sampling Freq : 10.1 Hz

> Hard CPU Time ZZ-Score (TPR/FPR) : 20.0 % / 0.0 %

> Hampel (TPR/FPR) : 20.0 % / 0.0 %

> Final Average (Z-Score) : 0.07

> Final Average (Hampel) : 0.07

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 7268 us

> End-to-End Latency : 2558 ms

> FFT Dominant Frequency : 5.04 Hz

eq: 16629 Hz

> CPU Time Z-Score : 382 us

> CPU Time Hampel : 557 us

> Z-Score (TPR/FPR) : 20.0 % / 0.0 %

> Hampel (TPR/FPR) : 20.0 % / 0.0 %

> Final Average (Z-Score) : 0.07

> Final Average (Hampel) : 0.07

SYSTE REPORT

> CPU Execution Time : 7268 us

> End-to-End Latency : 2558 ms

> FFT Dominant Frequency : 5.04 Hz

> eq: 16629 Hz

> CPU Time Z-Score : 382 us

> CPU Time Hampel : 557 us

> Z-Score (TPR/FPR) : 20.0 % / 0.0 %

> Hampel (TPR/FPR) : 20.0 % / 0.0 %

> Final Average (Z-Score) : 0.07

> Final Average (Hampel) : 0.07

SYSTEM PERFORMANCE REPORT

> CPU Execution Time : 7259 us

> End-to-End Latency : 2557 ms

> FFT Dominant Frequency : 5.02 Hz

> Adaptive Sampling Freq : 10.0 Hz

> Hardware Max Sampling Freq: 16629 Hz

> CPU Time Z-Score : 379 us

> CPU Time Hampel : 563 us

> Z-Score (TPR/FPR) : 36.4 % / 0.0 %

> Hampel (TPR/FPR) : 54.5 % / 0.0 %

> Final Average (Z-Score) : -0.13

> Final Average (Hampel) : -0.04

Analisi dell'Esperimento 3: Confronto segnale bonus filtrato e non filtrato

SYSTEM PERFORMANCE REPORT
> CPU Execution s
> End-to-End Latency : 1279 ms
> [Unfiltered] FFT Dominant Freq : 4.90 Hz
> [Unfiltered] Adaptive Samp Freq: 9.8 Hz
> [Z-SCR] FFT Dominant Freq : 4.86 Hz
> [Z-SCR] Adaptive Samp Freq: 9.7 Hz
> [HMPL] FFT Dominant Freq : 4.94 Hz
> [HMPL] Adaptive Samp Freq: 9.9 Hz
> CPU Time Z-Score : 224 us
> CPU Time Hampel : 282 us
re (TPR/FPR) : 25.0 % / 0.0 %
> Hampel (TPR/FPR) : 16.7 % / 0.0 %
> Final Average (Z-Score) : 0.04
> Final Average (Hampel) : -0.06

SYSTEM PERFORMANCE REPORT
> CPU Execution Time : 9368 us
> End-to-End Latency : 1279 ms
> [Unfiltered] FFT Dominant Freq : 4.90 Hz
> [Unfiltered] Adaptive Samp Freq: 9.8 Hz
> [Z-SCR] FFT Dominant Freq : 4.86 Hz
> [Z-SCR] Adaptive Samp Freq: 9.7 Hz
> [HMPL] FFT Dominant Freq : 4.94 Hz
> [HMPL] Adaptive Samp Freq: 9.9 Hz
> CPU Time Z-Score : 224 us
> CPU Time Hampel : 282 us
> Z-Score (TPR/FPR) : 25.0 % / 0.0 %
> Hampel (TPR/FPR) : 16.7 % / 0.0 %
> Final Average (Z-Score) : 0.04
> Final Average (Hampel) : -0.06

```
SYSTEM PERFORMANCE REPORT
> CPU Execution Time        : 9355 us
> End-to-End Latency        : 1279 ms
> [Req: 9.8 Hz
> [Z-SCR] FFT> [Z-SCR] Adaptive Samp Freq: 10.1 Hz
> [HMPL]  FFT Dominant Freq : 5.02 Hz
> [HMPL]  Adaptive Samp Freq: 10.0 Hz
> CPU Time Z-Score          : 225 us
> CPU Time Hampel           : 275 us
> Z-Score (TPR/FPR)         : 36.4 % / 0.0 %
> Hampel (TPR/FPR)          : 45.5 % / 0.0 %
> Final Average (Z-Score)   : -0.38
> Final Average (Hampel)    : -0.27
-----------------------------------

SYSTEM PERFORMANCE REPORT
> CPU Execution Time        : 9355 us
> End-to-End Latency        : 1279 ms
> [unfiltered]   FFT Dominant Freq : 4.89 Hz
> [unfiltered]   Adaptive Samp Freq: 9.8 Hz
> [Z-SCR] FFT Dominant Freq : 5.03 Hz
> [Z-SCR] Adaptive Samp Freq: 10.1 Hz
> [HMPL]  FFT Dominant Freq : 5.02 Hz
> [HMPL]  Adaptive Samp Freq: 10.0 Hz
> CPU Time Z-Score          : 225 us
> CPU Time Hampel           : 275 us
> Z-Score (TPR/FPR)         : 36.4 % / 0.0 %
> Hampel (TPR/FPR)          : 45.5 % / 0.0 %
> Final Average (Z-Score)   : -0.38
> Final Average (Hampel)    : -0.27
-----------------------------------

SYSTEM PERFORMANCE REPORT
> CPU Execution Time        : 9370 us
> End-to-End Latency        : 12582 ms
> [unfiltered]   FFT Dominant Freq : 4.97 Hz
> [unfiltered]   Adaptive Samp Freq: 9.9 Hz
p Freq: 9.9 Hz
> [HMPL]  FFT Dominant Freq : 49.55 Hz
> [HMPL]  Adaptive Samp Freq: 99.1 Hz
> CPU Time Z-Score          : 223 us
> CPU Time Hampel           : 278 us
> Z-Score (TPR/FPR)         : 18.8 % / 0.0 %
> Hampel (TPR/FPR)          : 18.8 % / 0.0 %
> Final Average (Z-Score)   : -0.13
> Final Average (Hampel)    : -0.13
-----------------------------------
```

SYSTEM PERFORMANCE REPORT
> CPU Execution Time : 9368 us
> End-to-End Latency : 1279 ms
> [unfiltered] FFT Dominant Freq : 5.02 Hz
> [unfiltered] Adaptive Samp Freq: 10.0 Hz
> [Z-SCR] FFT Dominant Freq : 5.05 Hz
> [Z-SCR] Adaptive Samp Freq: 10.1 Hz
> [HMPL] FFT Dominant Freq : 5.03 Hz
> [HMPL] Adaptive Samp Freq: 10.1 Hz
> CPU Timus
> Z-Score (TPR/FPR) : 15.4 % / 0.0 %
R) : 30.8 % / 0.0 %
> Final Average (Z-Score) : -0.02
> Final Average (Hampel) : -0.01

Bonus
SYSTEM PERFORMANCE REPORT
> CPU Execution Time : 9368 us
> FFT Dominant Freq : 5.02 Hz
> [unfiltered] Adaptive Samp Freq: 10.0 Hz
> [Z-SCR] FFT Dominant Freq : 5.05 Hz
> [Z-SCR] Adamp Freq: 10.1 Hz
> [HMPL] FFT Dominant Freq : 5.03 Hz
> Adaptive Samp Freq: 10.1 Hz
> CPU Time Z-Score : 226 us
> CPU Time Hampel : 277 us
> Z-Score (TPR/FPR) : 15.4 % / 0.0 %
> Hampel (TPR/FPR) : 30.8 % / 0.0 %
> Final Average (Z-Score) : -0.02
> Final Average (Hampel) : -0.01

> End-to-End Latency : 1279 ms
> [RAW] FFT Dominant Freq : 4.99 Hz
> [RAW] Adaptive Samp Freq: 10.0 Hz
> [Z-SCR] FFT Dominant Freq : 5.06 Hz
> [Z-SCR] Adaptive Samp Freq: 10.1 Hz
> [HMPL] FFT Dominant Freq : 5.08 Hz
> [HMPL] Adaptive Samp Freq: 10.2 Hz
> CPU Time Z-Score : 226 us
> CPU Ti% / 0.0 %
> Hampel (TPR/FPR) : 20.0 % / 0.0 %
> Final Average (Z-Score) : 0.15
> Final Average (Hampel) : 0.15

SYSTEM PERFORMANCE REPORT
> CPU Execution Time : 9374 us

```
> End-to-End Latency        : 1279 ms
> [RAW]   FFT Dominant Freq : 5.08 Hz
> [RAW]   Adaptive Samp Freq: 10.2 Hz
> [Z-SCR] FFT Dominant Freq : 5.10 Hz
> [Z-SCR] Adaptive Samp F5.15 Hz
> [HMPL]  Adaptive Samp Freq: 10.3 Hz
>          : 224 us
> CPU Time Hampel           : 283 us
> Z-Score (TPR/FPR)         : 25.0 % / 0.0 %
> Hampel (TPR/FPR)          : 25.0 % / 0.0 %
> Final Average (Z-Score)   : 0.07
> Final Average (Hampel)    : 0.07
-----------------------------------

SYSTEM PERFORMANCE REPORT
> CPU Execution Time        : 9374 us
> End-to-End Latency        : 1279 ms
> [RAW]   FFT Dominant Fr : 5.08 Hz
> [RAW]   Adaptive Samp Freq: 10.2 Hz
> [Z-SCR] FFT Dominant Freq : 5.10 Hz
> [Z-SCR] Adaptive Samp Freq: 10.2 Hz
> [HMPL]  FFT Dominant Freq : 5.15 Hz
> [HMPL]  Adaptive Samp Freq: 10.3 Hz
> CPU Time Z-Score          : 224 us
> CPU Time Hampel           : 283 us
> Z-Score (TPR/FPR)         : 25.0 % / 0.0 %
> Hampel (TPR/FPR)          : 25.0 % / 0.0 %
> Final Average (Z-Score)   : 0.07
> Final Average (Hampel)    : 0.07
-----------------------------------
```

Analisi Consumi

```
--------------------------------------------------
Battery (Load Voltage) : 0.92 V
ESP32 Voltage (Bus)    : 0.92 V
Current Consumed       : 42.00 mA
Instantaneous Power    : 36.00 mW
--------------------------------------------------
Battery (Load Voltage) : 0.88 V
ESP32 Voltage (Bus)    : 0.88 V
Current Consumed       : 41.70 mA
Instantaneous Power    : 36.00 mW
--------------------------------------------------
Battery (Load Voltage) : 0.87 V
```

```
ESP32 Voltage (Bus)    : 0.86 V
Current Consumed       : 42.60 mA
Instantaneous Power    : 36.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.94 V
ESP32 Voltage (Bus)    : 0.93 V
Current Consumed       : 105.50 mA
Instantaneous Power    : 92.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.91 V
ESP32 Voltage (Bus)    : 0.90 V
Current Consumed       : 105.80 mA
Instantaneous Power    : 92.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.88 V
ESP32 Voltage (Bus)    : 0.86 V
Current Consumed       : 132.80 mA
Instantaneous Power    : 112.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.88 V
ESP32 Voltage (Bus)    : 0.86 V
Current Consumed       : 134.50 mA
Instantaneous Power    : 114.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.87 V
ESP32 Voltage (Bus)    : 0.86 V
Current Consumed       : 105.50 mA
Instantaneous Power    : 92.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.87 V
ESP32 Voltage (Bus)    : 0.86 V
Current Consumed       : 105.50 mA
Instantaneous Power    : 92.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.94 V
ESP32 Voltage (Bus)    : 0.93 V
Current Consumed       : 105.70 mA
Instantaneous Power    : 92.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.94 V
ESP32 Voltage (Bus)    : 0.93 V
Current Consumed       : 105.90 mA
Instantaneous Power    : 92.00 mW
---------------------------------------------------
Battery (Load Voltage) : 0.94 V
ESP32 Voltage (Bus)    : 0.93 V
Current Consumed       : 107.40 mA
Instantaneous Power    : 92.00 mW
```